**In this question you must show all stages of your working.**

**Solutions relying entirely on calculator technology are not acceptable.**

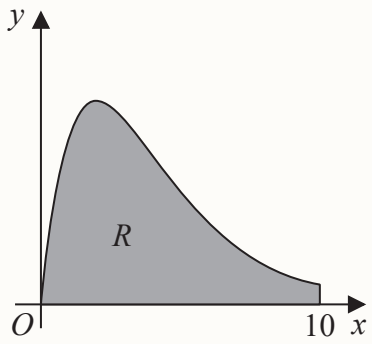



**Figure 2**

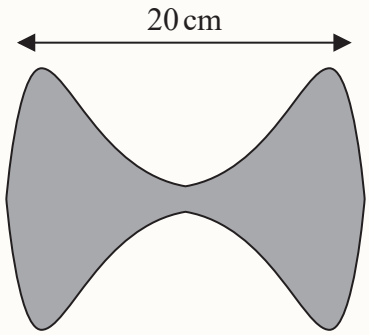



**Figure 3**

Figure 2 shows the curve with equation

$$y = 10x\mathrm{e}^{-\frac{1}{2}x} \qquad 0 \leqslant x \leqslant 10$$

The finite region $R$, shown shaded in Figure 2, is bounded by the curve, the $x$-axis and the line with equation $x = 10$

The region $R$ is rotated through $2\pi$ radians about the $x$-axis to form a solid of revolution.

(a) Show that the volume, $V$, of this solid is given by

$$V = k\int_0^{10} x^2\mathrm{e}^{-x}\,\mathrm{d}x$$

where $k$ is a constant to be found.

**(2)**

(b) Find $\int x^2\mathrm{e}^{-x}\,\mathrm{d}x$

**(3)**

Figure 3 represents an exercise weight formed by joining two of these solids together.

The exercise weight has mass 5 kg and is 20 cm long.

Given that

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

and using your answers to part (a) and part (b),

(c) find the density of this exercise weight. Give your answer in grams per $\text{cm}^3$ to 3 significant figures.

**(5)**